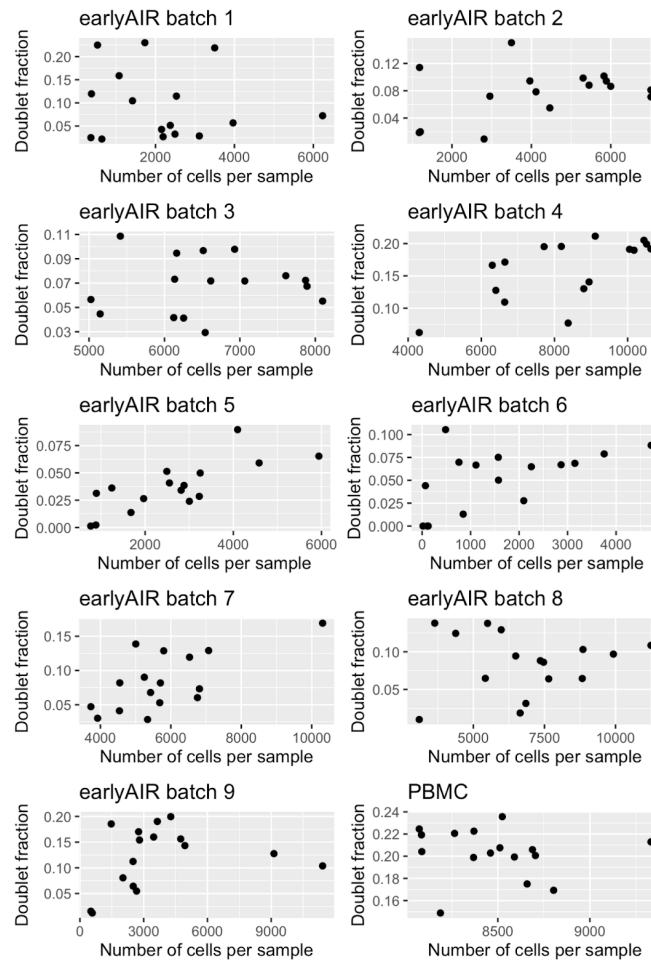
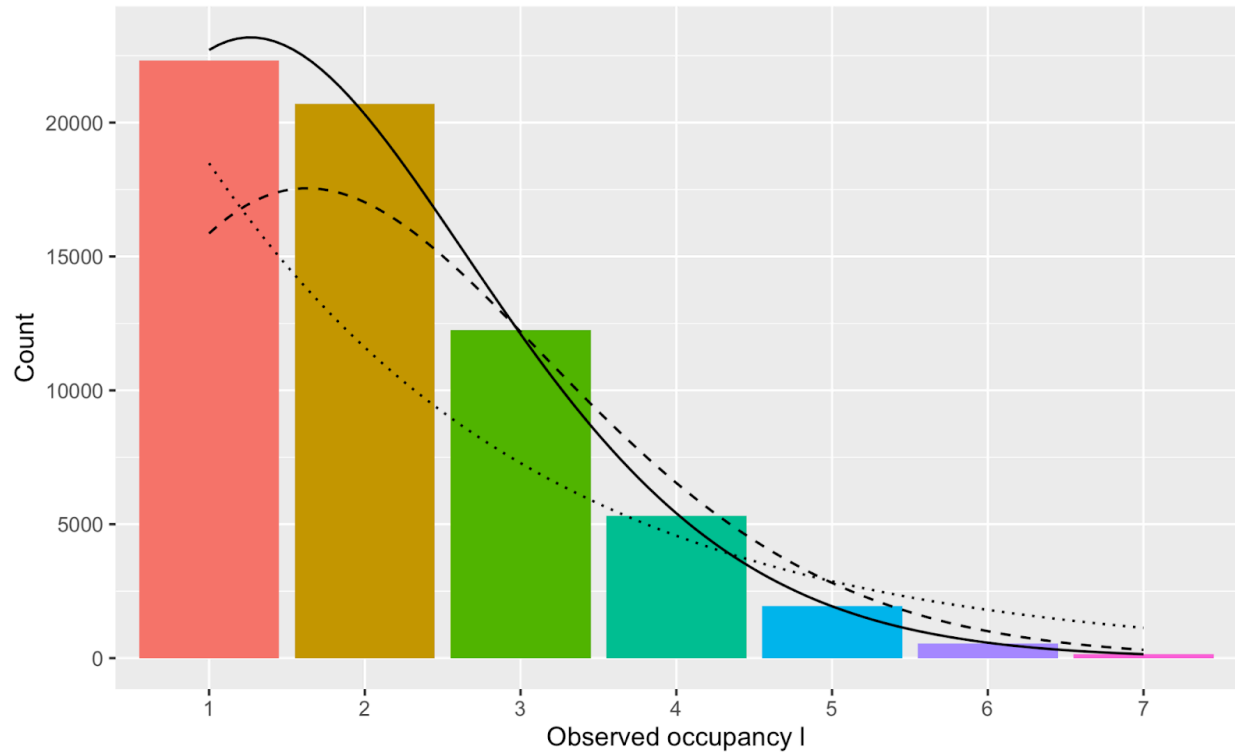


Supplementary materials

More cells, more doublets in highly multiplexed single-cell data

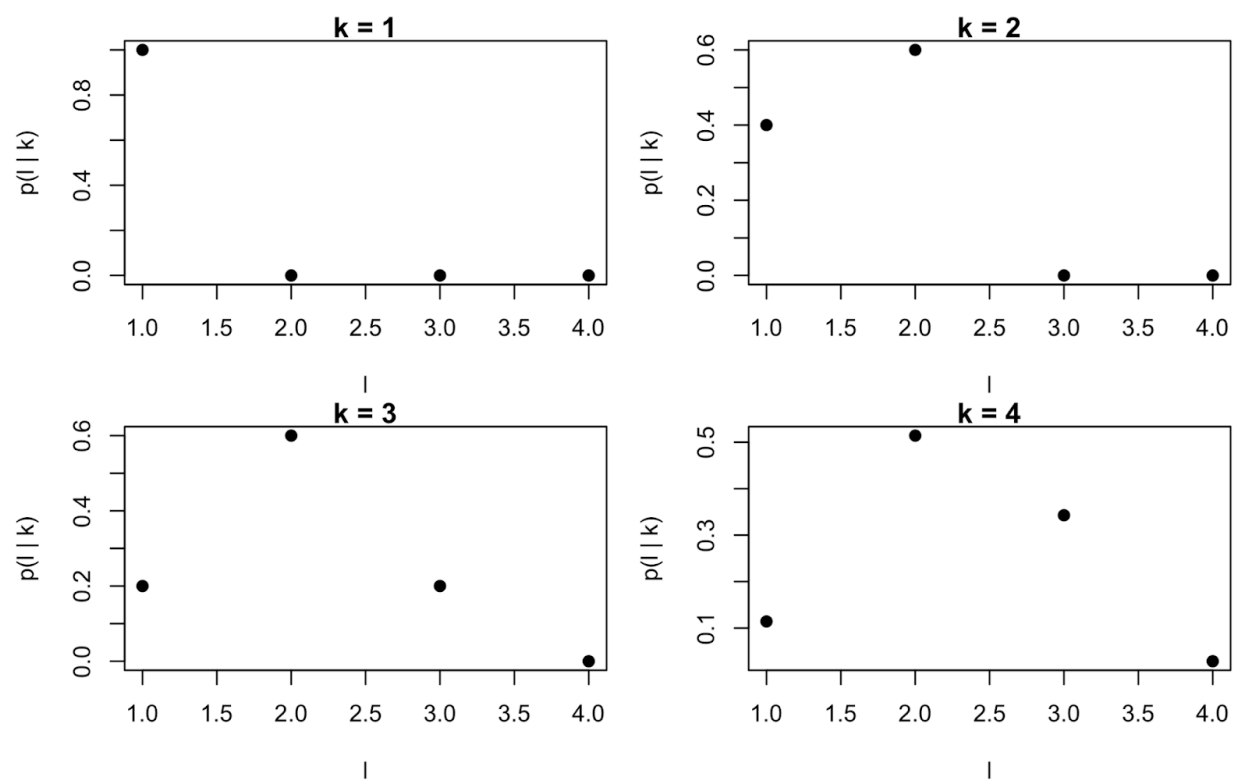


Supplementary Figure 1: Doublet fraction per sample vs number of cells for the earlyAIR and PBMC data sets.

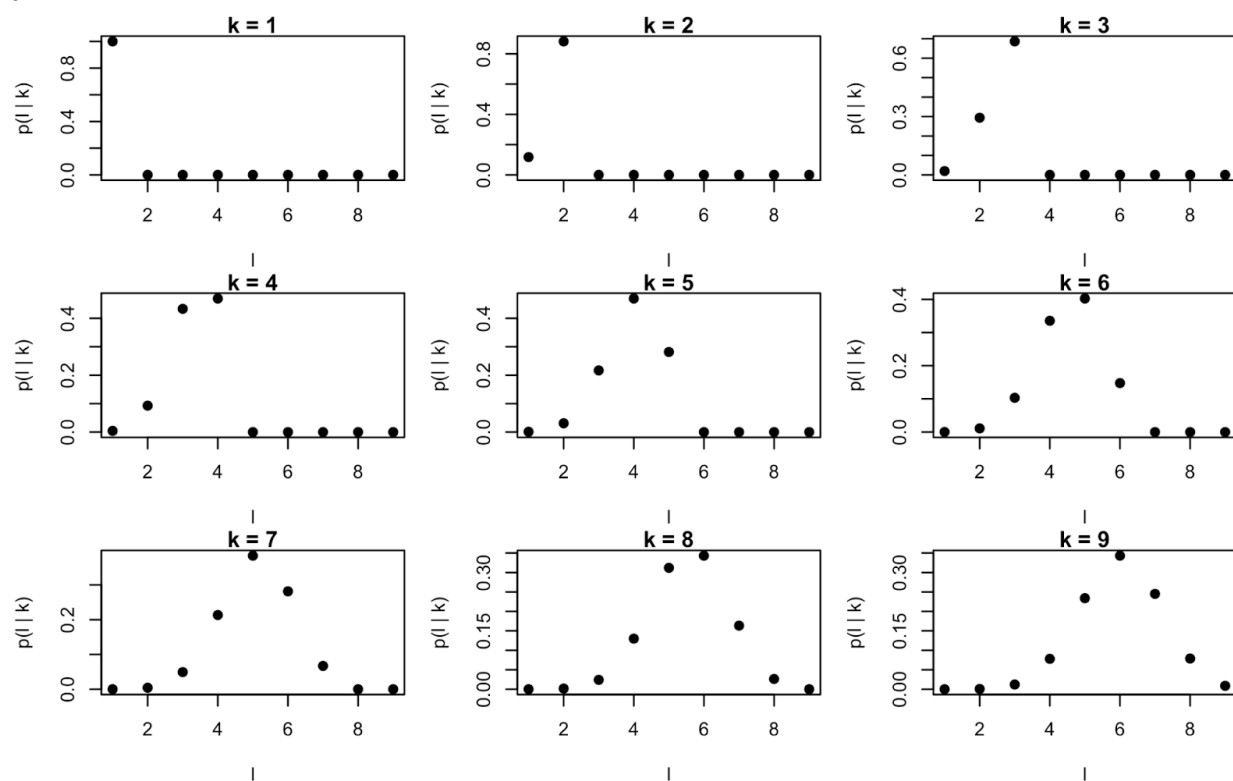


Supplementary Figure 2: Histogram showing the observed occupancy of droplets in the PBMC data set. Curves indicate fits to the data, with a Poisson fit (dashed line), zero-truncated Poisson fit (solid line) and exponential fit (dotted line).

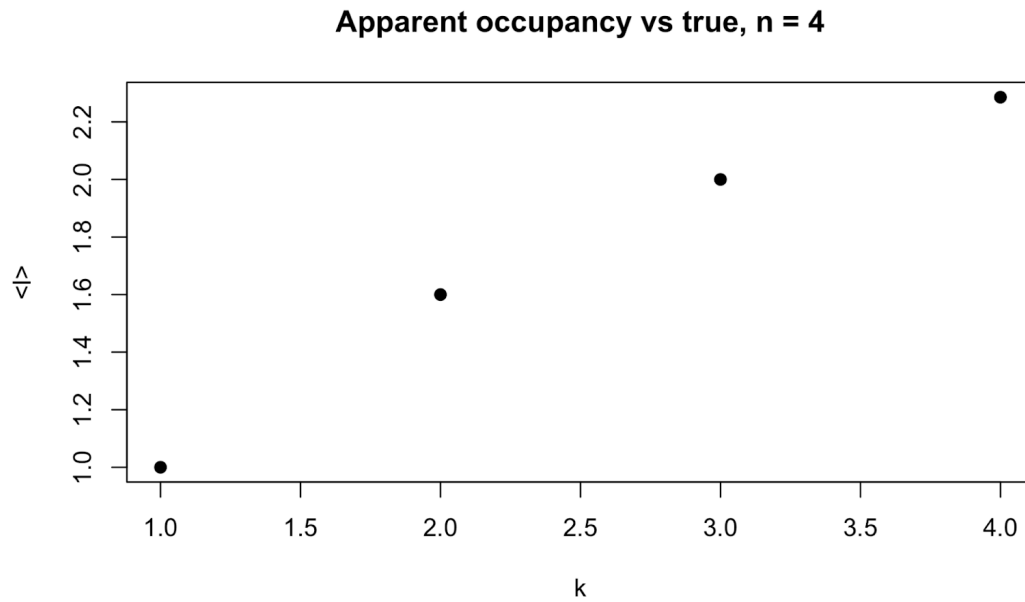
a



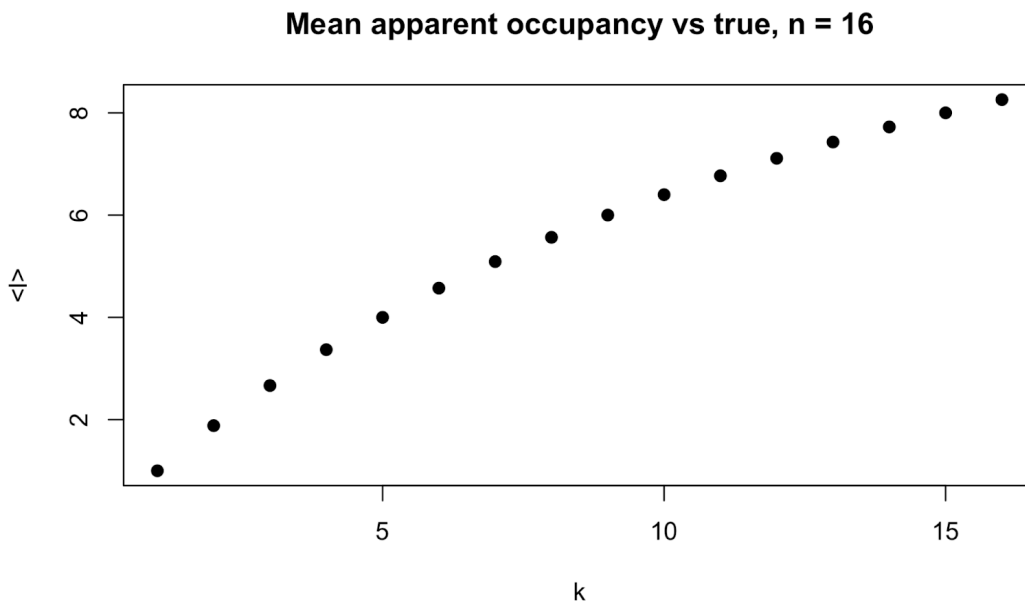
b



c



d



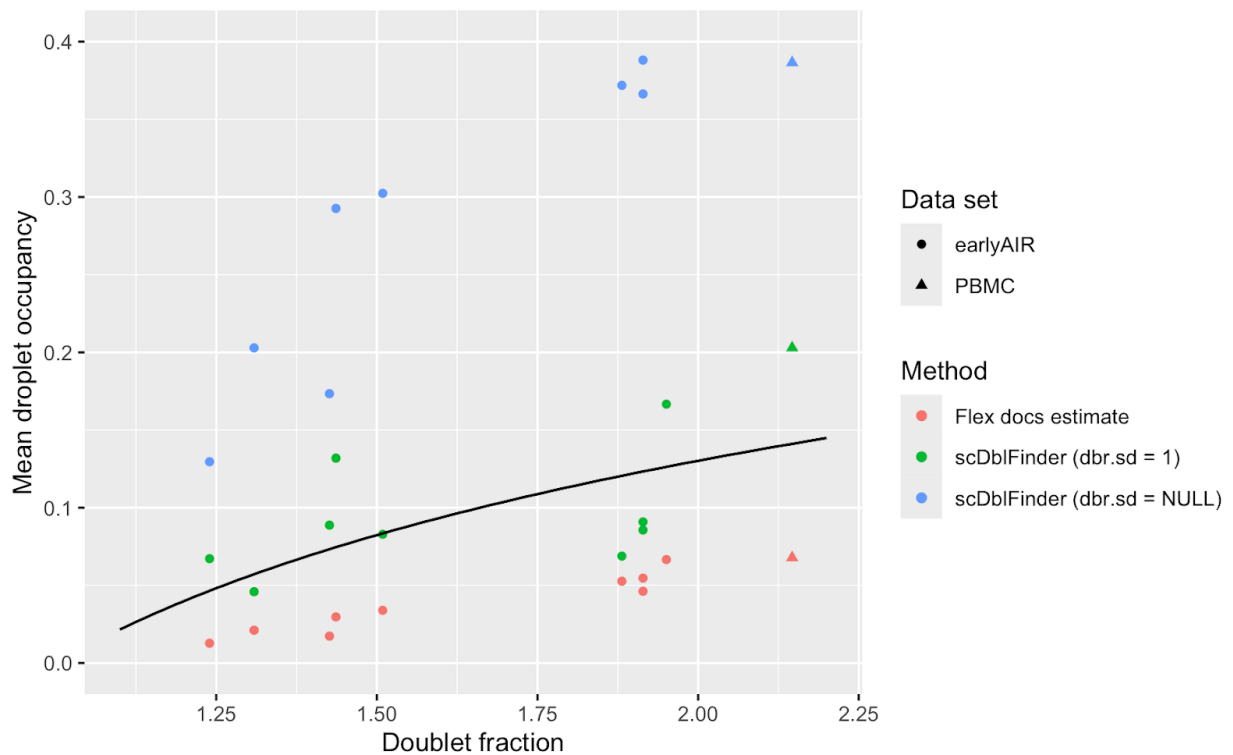
Supplementary Figure 3: Conditional probability of observing occupancy l given true occupancy k for a) $n = 4$ and b) $n = 16$. Mean observed occupancy $\langle l \rangle$ vs k for c) $n = 4$ and d) $n = 16$

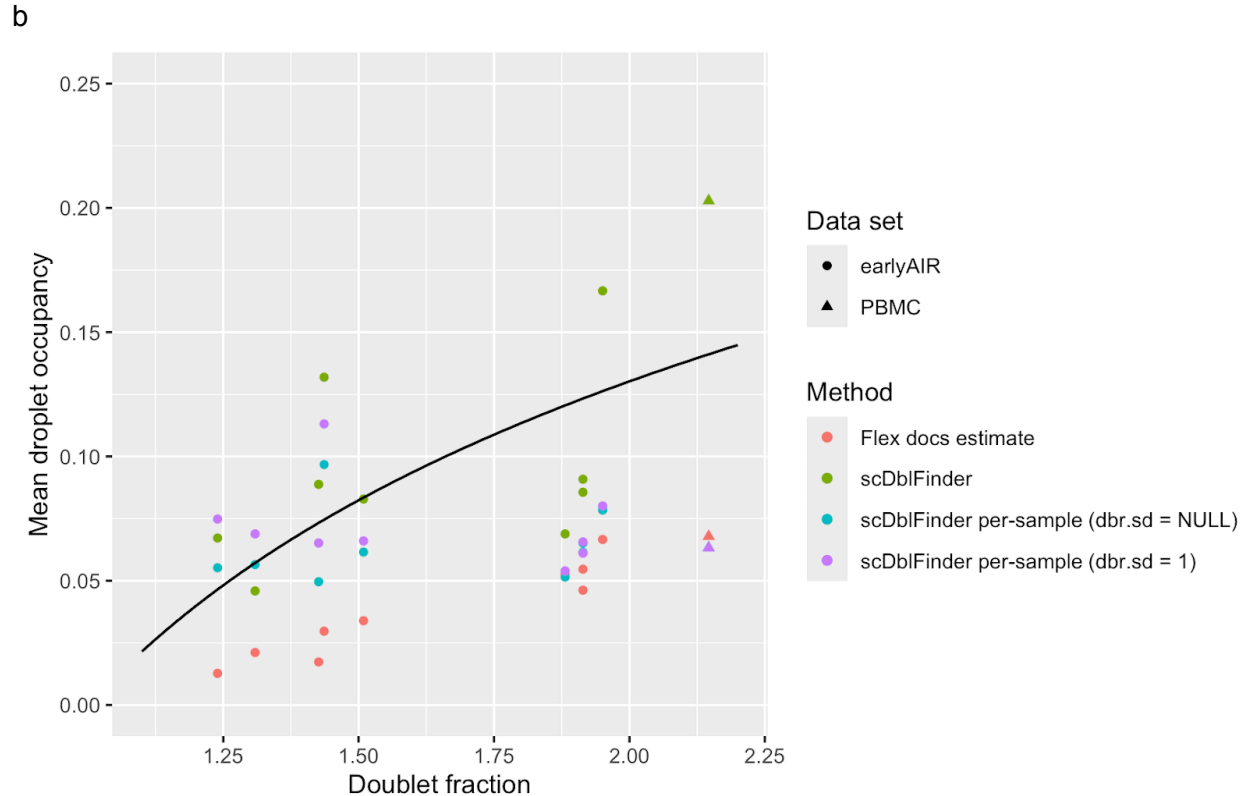
Testing scDbIFinder parameters

In an earlier version of this paper, the parameters used to run scDbIFinder were different to those in the current version. In particular, the option `dbr.sd = NULL` was used. With this option, scDbIFinder assumes a doublet rate in the data of 1% per 1000 cells, and uses this assumption to help determine the doublet score threshold for calling singlets or doublets. This rate is based on the expected rate from the 10X 3' protocol, and is not appropriate for Flex data, where the number of cells in a 16-plex experiment often exceeds 100,000.

In the current version of the paper, we have set `dbr.sd = 1`. This removes any assumption about the doublet rate and determines the threshold based on misclassification of artificial doublets in the classifier training stage. In our data sets, this results in lower overall doublet rates, and better agreement between the scDbIFinder calls and our theoretical model. We show the scDbIFinder results from both versions, along with the estimated values from the Flex documentation and our theoretical model in Supplementary Figure 4a.

a



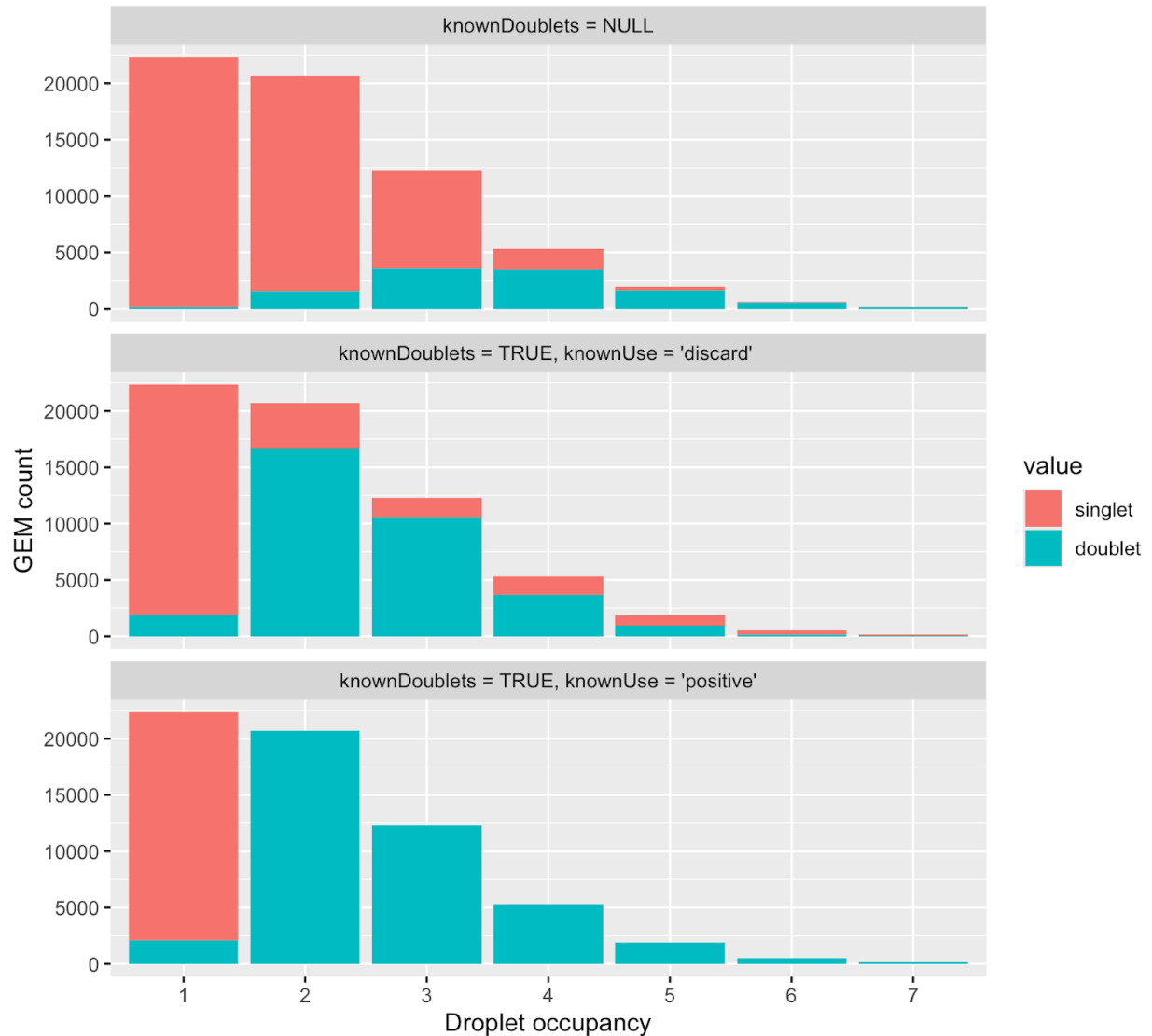


Supplementary Figure 4: Doublet fraction vs mean droplet occupancy using scDbIFinder with different parameter settings.

The `samples` argument in scDbIFinder allows the user to call doublets on subsets of the data. The intended use is for data which has been processed in separate captures, but we use it here to process each sample barcode separately, with both `dbr.sd = NULL` and `dbr.sd = 1`, the results are shown in Supplementary Figure 4b, along with the settings used in the main manuscript, the estimates from the 10X documentation and the theoretical curve.

Flex allows for a novel means of testing doublet finding directly using the RNA sequencing output. In a Flex experiment, due to superloading, many droplets contain multiple cells and cell calling is performed separately for each sample barcode. This means the initial step in a standard Flex analysis is to separate reads based on their combined droplet and sample barcode and then call cell-containing or empty “droplets” based on the data separated by samples. A droplet/sample combination may look empty even when there is a cell from a different sample in the same droplet. The resulting counts matrix has cells labeled by both the droplet barcode and sample barcode. Aggregating the counts for each “droplet” across the sample barcodes creates a droplet-level counts matrix analogous to 3’ scRNA-seq experiments. Since the number of cells (from unique samples) called by CellRanger in each droplet is known, the droplet-level counts matrix provides a ground truth data set for testing doublet finders: droplets with only one cell are singlets and droplets with two or more cells are doublets. We use the PBMC data set to briefly explore how different scDbIFinder settings can be used to optimise

doublet calling at the droplet level. The relevant parameter is `knownDoublets`, which allows a list of known doublets to be used for classification. A secondary option, `knownUse`, takes values of either `discard` or `positive`. In the first case, the list of known doublets is used only for determining the score threshold for calling doublets. In the latter, the known doublets are used in the training data set. We show in Supplementary Figure 5 the performance of scDbiFinder on the PBMC data set aggregated to the droplet level with three settings: 1) the same as used in the main manuscript; 2) `knownDoublets = TRUE` (a list of all droplets with occupancy > 1 is provided) and `knownUse = discard`; 3) `knownDoublets = TRUE`, `knownUse = positive`.



Supplementary Figure 5: Performance of scDbiFinder on droplet level counts with different parameter settings.

